Manufacturing and Processing Technology 2

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Manufacturing and Processing Technology 2 (A07957)

Short Title:	Manufac & Process Technology 2
Faculty	Science and Computing
Department	Science
Cluster	Pharmaceutical Science
Author	AHAYDEN
Credits	10

Level: Intermediate

Description of Module / Aims

This module will provide the student with an in-depth knowledge of the equipment utilised in manufacturing in the pharmaceutical and biopharmaceutical industries, how it operates as well as the processes themselves. The module describes in detail the scale-up issues and development cycle surrounding such processes. It also gives a basic introduction to aspects such as process control, risk assessment and process analytical technology (PAT).

Programmes

`MANF-0004`	BSc in Good Manufacturing Practice and Technology (`WD_SGMPT_D`)	1 / 1 / M

Indicative Content

- Process development, tasks and activities from discovery to technology transfer
- Extraction, chemical synthesis, semi-synthesis, biopharmaceutical production
- Pharmaceutical operations: review of facility design and plant layout; processing documentation
- Process control: a basic overview of the important components in a control system
- Chemical and Bio-reactors: design, control and operation, reaction kinetics
- Construction and operation of separation equipment: liquid-liquid extraction, distillation, filtration
- Drying, milling and powder properties
- Mechanisms of heat transfer, heat exchange equipment
- Down-stream biopharmaceutical processing; cleaning and sterile operation
- Aspects in scale-up and pilot plant production
- Quality by Design (QBD), Risk assessment and Process Analytical Technology (PAT)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Compare and contrast the operations involved in the synthesis and purification of (bio)pharmaceuticals on a large scale.
2. Illustrate and appraise the operation and control of processing equipment including (bio)reactors, liquid-liquid extractors, filters, dryers and milling equipment.
3. Examine the importance of design, layout and process control to (bio)pharmaceutical operations.
4. Examine critical milestones and pertinent issues surrounding chemical process development and scale-up.
5. Analyse and classify risk and learn how to reduce risk through process understanding and experimental design.
6. Research individual processes and their underlying theory.
7. Complete and record laboratory experiments to establish pharmaceutical structure and chirality.

Learning and Teaching Methods

- Lectures.
- E-learning via Moodle.
- Visits to nominated pharmaceutical companies.
- Set problems.
- Written assignments and in-class discussion.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
Lab Report	30%	7
Assignment	20%	5,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Lab		12
Independent Learning		234

Supplementary Material

- Anderson, N. _Practical Process Research and Development_. Oregon: Academic Press, 2012.

- Lee, S. and G. Robinson. _Process development:fine chemicals from grams to kilograms_. UK: Oxford University Press, 1995.
- Moulijn, J.A., M. Makkee and A.E. Van Diepen. _Chemical Process Technology_. Netherlands: J. Wiley & Sons, 2013.
- Nusim , S. _Active Pharmaceutical Ingredients: Development, Manufacturing, and Regulation (Drugs and the Pharmaceutical Sciences)_. Edition. New York: Informa Healthcare, 2009.

Requested Resources

- Room Type: Chemistry Lab